

Muhammad Saram Hassan
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EDUCATION

2022-Present Lahore University of Management Sciences - BSc. Computer Science
BS in Computer Science
Key Courses: Advanced Topics in Machine Learning[†], Distributed Systems[†], Operating Systems, Linear Algebra
[†] Graduate Level courses

Honors and Awards

- ❖ Placed on the Dean's Honors List
- ❖ Silver Medalist - International Youth Mathematical Championship amongst 12500+ participants
- ❖ Ranked Top 20 at Pakistan National Mathematics Talent Contest (equiv. USAMO for IMO)
- ❖ 1st Place – Pakistan National Engineering and Robotics Contest at NUST
- ❖ 1st Place - International Kangaroo Linguistics Contest, Amongst 3000+ competitors worldwide, recognized for exceptional linguistic problem-solving.
- ❖ Ranked in Top 30 with an Honorable Mention at CERN BL4S Particle Physics Competition

PROJECTS AND RESEARCH

Federated Learning Project

- Developed and rigorously implemented a suite of federated learning algorithms—FedAverage, FedSGD, FedSAM, FEDGH, and SCAFFOLD—by adapting insights from cutting-edge research literature to tackle diverse, real-world data settings.
- Conducted empirical evaluations across datasets with Dirichlet-induced label imbalances (levels 0.1, 0.5, and 2), effectively simulating a spectrum of data heterogeneity from mild to extreme.

Exploring Model Compression and Pruning Techniques through implementation (MIT TinyML Course)

- Leveraged structured and unstructured pruning techniques on the VGG16 model, building on the foundational principles from the MIT TinyML course and "Optimal Brain Damage" methodology.
- Applied an L2-norm-based unstructured pruning method, eliminating weights with minimal contributions to performance. Conducted structured channel-wise pruning for comparative analysis, adjusting sparsity ratios to achieve a balanced reduction in network complexity.
- Performed sensitivity analysis on individual layers to identify pruning targets with minimal impact on accuracy. Pruned select layers to achieve a 75% compression, prioritizing computational efficiency.
- Insights:
 - Unstructured Pruning: Achieved a 65.9% accuracy after fine-tuning, a 7.52% improvement over the original unpruned model.
 - Structured Pruning: Channel-wise pruning increased accuracy by 3% after fine-tuning. too
 - Observed that weights exhibited nearly equal magnitudes, in later layers complicating importance ranking by L2-norm.

Research Experience

Undergraduate Research Assistant

Department of Electrical Engineering,

Advisor: Muhammad Tahir

October 24– Present

- Conducting a comparative analysis of pruning techniques for neural networks, focusing on VGG-16, Resnet and ViT architecture and CIFAR-100 dataset using Taylor approximations of loss and similarity metrics to evaluate computational efficiency and model performance.
- Exploring out-of-distribution generalization with style-transferred datasets to analyze shape vs. texture bias in pruned models with the aim to uncover key reasons behind the performance differences across pruning techniques and provide actionable insights for efficient model compression in resource-constrained environments.

Integration of Large Language Models in Mobile Banking for Regional Languages.

Feb. 2024 – Present

- To integrate multilingual language models into mobile banking platforms, enabling support for Urdu, Punjabi, and Siraiki languages with accent recognition capabilities.
- Led a team of 50+ LUMS and high school students in collaborating with Uplift AI, a US-based AI startup, and developed and tested modules for accurate interaction tracking and user feedback analysis, focusing on regional accents.
- Status: Testing phase underway, with plans to refine the application based on user feedback before scaling deployment.

Distributed Systems

Raft Algorithm for Fault-Tolerant Key-Value Storage

- Implemented the Raft algorithm in Golang to build a fault-tolerant key-value storage service, ensuring consistency across replicas via log consensus and leader election mechanisms.

Multithreading and Concurrency Control in C

- Developed threading primitives and scheduling mechanisms for thread lifecycle management. Implemented synchronization techniques (e.g., mutex locks, condition variables) to ensure thread-safe operations.

Chrono Savior - 2D Adventure Game Development

- Developed a time-travel-themed 2D game incorporating shooter, action, and adventure genres using Unity with 3 other people.
- Designed core mechanics such as player movement, space combat, and time-based abilities and developed modular game modes including Infinite Mode, Space Travel, and Ground Combat.

LEADERSHIP & EXTRACURRICULARS

LUMS Entrepreneurial Society (LES)

Project Head for Social Outreach Program.

2024-Present

Developing the curriculum for Shaur Summer internship for high school interns whereby modules such as basic game development, usage of AI in Social Entrepreneurship & critical thinking, and money management

Assistant Director IT & Automations

2023-2024

- Designed and developed comprehensive WordPress websites for LES, YLES Projects, and Initiate, enhancing their online presence and functionality.
- Built an Excel simulation that deployed a framework of Strategic Business Management, analyzing the effects of different strategies on key factors such as MR growth, CAC reduction, and CLV growth.
- Utilized the capabilities of Excel, Python, WordPress, and Google Forms to simulate the intricacies of the real stock market and introduce the financial world to 500+ high school students at LUMS Young Leaders Entrepreneurs Summit.
- Ensured technical growth within the society by conducting various IT bootcamps

LUMS Society for Promotion & Development of Engineering & Science (SPADES)

Lead Research Ventures

2024-Present

- Worked on the prototype for Pakistan's first student-made propellant rocket to reach a height of 1 kilometer; Team Lead for 12 LUMS students handling the Avionics, Propellants and Design of the rocket.

Director Observatory

2023-2024

- Revamped the LUMS observatory by fixing the AVX of Celestron 6SE; organized 5+ interactive workshops on astronomy, astrophysics and astrophotography for the community `

Technical Skills

Languages, Libraries, and Frameworks: Python, C, C#, unity, C++, NumPy, JAX, Pandas, Matplotlib, Seaborn, TensorFlow, PyTorch, Scikit-Learn, Golang

Skills and Other Tools: Linear Algebra, Google Colab/Jupyter Notebooks, Computer Vision, NLP, Machine/Deep Learning, Calculus, MATLAB and elementary signal processing, Elementary Frontend Development